Chapter 1
The International System of Units (SI)



1.1 Units of Measure: A Brief History

Since the beginning of civilization, human beings have always had to perform measurements, even if this was done intuitively. When they began to live in a community, this need to measure and weigh increased. With the advent of commerce and the establishment of private property, measuring the land and the results of its work was necessary for sale or exchange.

The first units of measure arose from their daily use and were based on parts of the human body. In principle, they could be considered "references," that is, anyone else could verify a measure. Thus emerged measurement units such as the inch, the hand, the foot, the yard, and the step. It is obvious these "references" were not fixed, as the human body is not standardized, and measures vary from individual to individual.

The Egyptians also used the size of the cubit, one of the forearm bones, as a standard of length measurement. Again, as the cubit varied from one person to another, Pharaoh Khufu established a granite standard based on the bone length of your arm during the construction of its pyramid (about 2900 BC). This pattern, whose reproduction we see in Fig. 1.1, was called the Egyptian royal cubit.

Over time, wooden bars to facilitate transportation replaced the granite bars, but as wood was worn out, lengths equivalent to the royal cubit were recorded on the walls of the main temples. This way, people could periodically check their wooden bars or do others.

In France, in the seventeenth century, a unit of linear measurement was standardized in a two-pin iron bar at the extremes, forming a calibrator. The distance between these two pins was considered a "*toise*," the bar was spoiled on the outer wall of the Grand Châtelet, the fortification that kept the head of one of the bridges of access to Paris.

Thus, as in the case of the standard cubit, interested parties could check their measuring instruments (Fig. 1.2).

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promulgated Imperial Law Number 1157 and adopted the French decimal metric system throughout the country. As mentioned earlier, Brazil was one of the first nations to adopt the new system.

In 1961, the National Institute of Weights and Measures (INPM) was created, and the International System of Units became the official system through Decree 52,243 of August 30, 1963, later replaced by Decree 63,323, September 12, 1968. In 1973, the INPM was replaced by the National Institute of Metrology, Standardization and Industrial Quality (INMETRO), currently called the National Institute of Metrology, Quality and Technology.

1.2 The International System of Units (SI)

The International Vocabulary of Metrology (VIM) defines the International System of Units (SI) as follows:

> System of units, based on the International System of Quantities, their names and symbols, including a series of prefixes and their names and symbols, together with rules for their use, adopted by the General Conference on Weights and Measures (CGPM). [VIM—1.16]

To complete the reasoning, we present the definition of the International System of Quantities:

> System of quantities based on the seven base quantities: length, mass, time, electric current, thermodynamic temperature, amount of substance, and luminous intensity.[VIM—1.6]

Some characteristics of SI:

- Unique units which can be reproduced and performed anywhere in the world;
- Few base units, separate and independent;
- Coherent, the combination of existing units produces other units without constants.

Knowing a Little More…
Since 1970, the BIPM has published the International System of Units in *SI Brochure* or *Brochure Sur Le SI* (in French), printed and digital versions. (source: https://www.bipm.org/en/publications/si-brochure)

The SI is the system of units in which:

- The unperturbed ground state hyperfine transition frequency of the cesium 133 atom $\Delta\nu_{Cs}$ is 9,192,631,770 Hz,
- The speed of light in vacuum c is 299,792,458 m/s,
- The Planck constant h is 6.626,070 15 × 10^{-34} J s,
- The elementary charge e is 1.602,176 634 × 10^{-19} C,
- The Boltzmann constant k is 1.380,649 × 10^{-23} J/K,
- The Avogadro constant N_A is 6.022,140 76 × 10^{23} mol^{-1},

- The luminous efficacy of monochromatic radiation of frequency 540×10^{12} Hz, K_{cd}, is 683 lm/W,

  where the hertz, joule, coulomb, lumen, and watt, with unit symbols Hz, J, C, lm, and W, respectively, are related to the units second, meter, kilogram, ampere, kelvin, mole, and candela, with unit symbols s, m, kg, A, K, mol, and cd, respectively, according to $Hz = s^{-1}$, $J = kg\ m^2\ s^{-2}$, $C = A\ s$, $lm = cd\ m^2\ m^{-2} = cd\ sr$, and $W = kg\ m^2\ s^{-3}$.

The numerical values of the seven defining constants do not have uncertainty. (Source: SI brochure 9th edition)

1.2.1 Quantity

Property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed as a number and a reference.

A reference can be a measurement unit, procedure, reference material, or a combination.

The concept of 'quantity' may be generically divided into, e.g., 'physical quantity,' 'chemical quantity,' and 'biological quantity,' or base quantity and derived quantity. [VIM—1.1]

Based on the information found in VIM, the nature of quantity is a common aspect of mutually comparable quantities. The division of "quantity" according to the "nature of a quantity" is somewhat arbitrary.

Knowing a Little More…
The quantities diameter, circumference, and wavelength are usually considered of the same nature, that is, of the quantity called length.

Heat, kinetic energy, and potential energy are usually considered part of the same quantity called energy.

Quantities of the exact nature in a given system of quantities have the same dimension. However, quantities of the same dimension are not necessarily similar.

The momentum of strength and energy are not, by convention, considered of the exact nature, although they have the same dimension. The same occurs for thermal capacity and entropy, as well as for the number of entities, relative permeability, and mass fraction.

A base quantity is the greatness of a chosen subset by convention of a given system of quantities, in which no greatness of the subgroup can be expressed as a function of others. The subset mentioned in the definition is called a set of base quantities. Base quantities are considered mutually independent, as a product of powers from other base quantities cannot express base quantities.

A derived quantity is defined as a function of the base quantities of this system.

In a system of quantities with the length and mass as its base quantities, the specific mass is a derived quantity defined by the quotient of a mass by a volume (length to the cube).

1.2.2 Measurement Unit

Real scalar quantity, defined and adopted by convention, with which any other quantity of the same kind can be compared to express the ratio of the two quantities as a number.
NOTE 1 Measurement units are designated by conventionally assigned names and symbols.
NOTE 2 Measurement units of quantities of the same quantity dimension may be designated by the same name and symbol even when the quantities are not of the same kind. For example, joule per kelvin and J/K are respectively the name and symbol of both a measurement unit of heat capacity and a measurement unit of entropy, which are generally not considered to be quantities of the same kind. However, in some cases, special measurement unit names are restricted to be used with quantities of a specific kind only. For example, the measurement unit 'second to the power minus one' (1/s) is called hertz (Hz) when used for frequencies and becquerel (Bq) when used for activities of radionuclides.
NOTE 3 Measurement units of quantities of dimension one are numbers. In some cases, these measurement units are given special names, e.g., radian, steradian, and decibel, or are expressed by quotients such as millimole per mole equal to 10^{-3} and microgram per kilogram equal to 10^{-9}.
NOTE 4 For a given quantity, the short-term "unit" is often combined with the quantity name, such as "mass unit" or "unit of mass." [VIM—1.9]

In SI, there are two classes of measurement units: the base and the derived units.

1.2.3 Base Unit

Measurement unit that is adopted by convention for a base quantity. [VIM—1.10]

Base units are seven independent physical quantities. Table 1.2 presents the new definitions and symbols of the base units since May 20, 2019.

Knowing a Little More …
The General Conference on Weights and Measures (CGPM), in its 25th meeting in November 2014, adopted a resolution on the new revision of the International Unit System (SI), validated at the 26th meeting in 2018. In this review, the kilogram, ampere, kelvin, and mole were redefined based on the fixed numerical values of Planck constant (h), elementary load on a proton (e), Boltzmann constant (k), and Avogadro constant (N_A), respectively.

Subsequently, the seven basic SI units were defined based on seven reference constants, to be known as "SI defining constants": the hyperfine transition frequency of the cesium—second; the speed of light in the vacuum—meter; the Planck constant—kilogram; the elementary load in a proton—ampere; Boltzmann's constant—kelvin; the Avogadro constant—mol; and the luminous efficacy of a specified monochrome source—candela. This has resulted in a more straightforward and more fundamental definition of the

(continued)

whole and dismissed the last of the definitions based on a material artifact—the international prototype of the kilogram maintained in BIPM.

The main disadvantage of the old definition of the kilogram was that it referred to the mass of the artifact, which, by nature, we know is not stable. Comparisons between the official copies and the international prototype showed some disagreement over time. The drift in the mass of the international prototype, since 1889 has yet to be demonstrated, but it should undoubtedly be present. The change rate of its mass could be determined only by insufficiently high absolute experiences.

The new kilogram unit can measure with the "watt scale (or kibble scale)," an instrument that allows mechanical with electromagnetic energy to compare two separate experiences. Ampere can be measured using Ohm's law ($A = V/\Omega$) and practical achievements of V and Ω, based on Josephson and Quantum Hall effects. Kelvin can be defined by the new system using acoustic thermometry. The technique allows you to determine the speed of sound in a sphere full of gas at a fixed temperature. The mole can be performed as the accurate amount of atoms in a perfect sphere of pure-28 silicon. (Source: Adapted from https://www.bipm.org/en/publications/misses-en-pratique/)

1.2.4 Derived Unit

Measurement unit for a derived quantity. [VIM—1.11]

They are units formed by combining base units according to mathematical relations that correlate the corresponding quantities. Table 1.3 presents some examples of derived units.

1.2.5 Dimensional Analysis of the Quantities

Dimensional analysis studies the quantities and relationships between the respective measurement units of these quantities. The study of dimensional analysis becomes a powerful ally that helps us write SI and obtain some equations involving physical quantities.

The VIM provides the following definition for quantity dimension:

Expression of the dependence of a quantity on the base quantities of a system of quantities as a product of powers of factors corresponding to the base quantities, omitting any numerical factor. [VIM—1.7]

Table 1.4 presents the symbols corresponding to the dimensions of quantities.

According to VIM—1.7, the size of a quantity Q is represented by:

$$\dim(P) = \frac{\dim(\text{force})}{\dim(\text{area})}$$

$$\dim(P) = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

The dimension of v^2 is:

$$\dim\left(v^2\right) = L^2T^{-2}$$

Therefore, the dimension of k is:

$$\dim(k) = \frac{\dim(P)}{\dim(v^2)} = \frac{ML^{-1}T^{-2}}{L^2T^{-2}} = ML^{-3}$$

In SI, this represents, $Un(k) = \frac{\text{kg}}{\text{m}^3}$, the density unit. Therefore, the correct answer is (b).

1.2.6 Decimal Multiples and Submultiples

Multiples and submultiples were defined in the SI, with the names and symbols given in Table 1.5.

Except for the prefixes da (deca), h (hecto), and k (kilo), all multiple prefix symbols are written with capital letters, and all submultiple symbols are written with lowercase letters. All prefix names are written with lowercase letters, except at the beginning of a sentence.

Table 1.5 Decimal multiples and submultiples

Factor	Prefix	Symbol	Factor	Prefix	Symbol
10^{30}	ronna	R	10^{-1}	deci	d
10^{27}	quetta	Q	10^{-2}	centi	c
10^{24}	yotta	Y	10^{-3}	milli	m
10^{21}	zetta	Z	10^{-6}	micro	μ
10^{18}	exa	E	10^{-9}	nano	n
10^{15}	peta	P	10^{-12}	pico	p
10^{12}	tera	T	10^{-15}	femto	f
10^{9}	giga	G	10^{-18}	atto	a
10^{6}	mega	M	10^{-21}	zepto	z
10^{3}	kilo	k	10^{-24}	yocto	y
10^{2}	hecto	h	10^{-27}	ronto	r
10^{1}	deca	da	10^{-30}	quecto	q

Source: SI brochure 9th edition

Although the multiples da (deca) and h (hecto) and submultiple d (deci) are not foreseen, their use is not shared, and it is recommended to express in k (kilo), m (milli), or μ (micro).

1.2.7 SI Units and Symbols Writing Rules

The writing rules of symbols and units were initially proposed by the 9th CGPM in 1948. They were then adopted by ISO/TC 12 (ISO 31, quantities and units). Some rules are presented below.

1. The symbols are expressed with lowercase letters and in Roman characters.

   Example: meter (m) second (s)
   The exceptions are the Greek letter Ω (unit of electrical resistance) and the liter unit, which can also be written with L.
   Note: The liter is not an SI unit, but its use is accepted.

2. If the unit's name is a proper name, the first letter of the symbol is capitalized, but it is written with a lowercase letter.

   Example: pascal (Pa) kelvin (K)
   The spelling of °C is degree Celsius, as the degree unit begins with a lowercase letter. Celsius is an adjective that starts with a capital letter because it is a proper name.

3. The symbols of the units have no plural and are not followed by points.

   Example: 10 kg 500 m 25 s

4. When dividing one unit by another, use an inclined bar, horizontal trace, or negative power.

   Example: km/h $\frac{\text{km}}{\text{h}}$ km h^{-1}

5. To avoid ambiguities, use only one inclined bar, parentheses, or negative powers.

   Example: m/s^2 or m s^{-2} and never m/s/s

6. The multiplication of the symbols of the units must be indicated by a space or a point centered at half height (·).

   Example: newton meter → N m or N•m

7. The tonic accent does not fall on the prefix but on the unit.

   Example: micro**me**ter kilo**me**ter

8. In writing a unit composed of the multiplication of unit names, a space or a hyphen should be used to separate the names from the units.

Example: Pa s → pascal second or pascal-second.

9. Time measurements:

   Correct: 5 h 14 min 3 h 30 min 15 s 2 h
   Wrong: 5:14 h 3 h 30′15′ 3:30:15 h
   Note: The hour (h) and the minute (min) are not SI units, but their use is accepted.

10. The numerical value precedes the unit, and there is always a space between the number and the unit. Thus, since the value of a quantity is the product of a number by a unit, the space is considered a sign of multiplication.

    Example: 124.6 mm 45.9 °C 50 kg
    Exception for this rule is the symbols of the units of grade (°), minute (′), and second (″) of the flat angle (units outside of SI), for which there is no space between the numerical value and the symbol of the unit. Example: 45° 25′ 6″

11. When a multiple or submultiple prefix is used, it is part of the unit and precedes the symbol of unity without space between the prefix symbol and the unit symbol.

    Correct: 124.6 mm (numerical value/space/prefix of the unit/unit)
    Wrong: 124.6 m m

12. Do not mix the name with the symbol.

    Correct: kilometer per hour or km/h
    Wrong: km/hour or kilometer/h

1.2.8 Non-SI Units Accepted for Use with the SI

BIPM recognizes the need to use widely used units, although they are not part of the SI. Table 1.6 presents some of these units.

1.3 Proposed Exercises

1.3.1 What is the symbol of the quantity length in SI?

(a) mts
(b) m
(c) KM
(d) km